

AI for Bharat Hackathon

Powered by **aws**



Team Name : Future Forward India

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Problem Statement 3: Build an AI-powered solution that improves access to information, resources, or opportunities for communities and public systems.

Jan Sahayak – AI Powered Government Scheme Eligibility & Guidance Platform

The Problem Statement:

India spends ₹14+ lakh crore annually on 2,000+ welfare schemes, yet an estimated 60% of eligible citizens never access their entitled benefits.

The reasons are clear:

- **Language Barrier** – Only 13% of Indians are comfortable with English; most government portals are English/Hindi only
- **Awareness Gap** – 60% of rural Indians are unaware of schemes they qualify for (NCAER Study)
- **Complexity** – Each scheme has different eligibility criteria, documents, and application processes
- **Digital Divide** – Only 50 crore of 130 crore Indians use smartphones; form-filling is impossible for many
- **Result:** ₹5.6 lakh crore in welfare funds go unclaimed every year. A farmer in Jharkhand loses ₹18,000/year in PM-KISAN simply because he didn't know the scheme existed.

Our Solution – Jan Sahayak:

An AI-powered, voice-first, multilingual web application that enables any Indian citizen to discover government schemes they are eligible for – by simply having a conversation in their language or uploading a photo of their documents. No forms. No digital literacy required. Just speak and discover your rights.

1. How different is it from existing solutions?

Existing Solution	What It Does	Why It Falls Short	Jan Sahayak Advantage
MyScheme.gov.in	Form-based eligibility checker	20+ field forms, English/Hindi only, requires digital literacy	Conversational AI – just talk, no forms needed
UMANG App	Government services app	Requires app download, complex navigation	Browser-based, zero download, voice-first
CSC Centers	Physical service delivery	Requires physical visit, long queues, limited hours	Access from home, 24/7, instant response
ChatGPT/Generic AI	General Q&A	No verified scheme database, hallucination risk, English-centric	Curated database of 30+ verified schemes, zero hallucination on eligibility
Google Search	Information search	Scattered, not personalized, overwhelming	Personalized to YOUR profile, ranked by relevance

2. How will it solve the problem?

1. **Eliminates Language Barrier** → Speak in Hindi, Tamil, Telugu, Marathi or any Indian language
2. **Eliminates Awareness Gap** → AI proactively discovers ALL schemes you're eligible for
3. **Simplifies Complexity** → AI handles eligibility logic; user just answers simple questions
4. **Bridges Digital Divide** → Voice-first design; no reading, typing, or form-filling required
5. **Makes Action Clear** → Document Readiness Score tells you exactly what to do next

3. USP of the Proposed Solution

Innovation 1: Document-to-Eligibility Pipeline — Upload a photo of your Aadhaar card → AI extracts your details using OCR + LLM → Instantly calculates eligibility across 30+ schemes. No existing solution does this.

Innovation 2: Document Readiness Scoring — For each scheme, Jan Sahayak shows: "You have 3/5 documents ready. Missing: Land Record (get from Patwari, 3-7 days) and Income Certificate (get from Collectorate, 7-15 days)." No existing solution tells you the gap between what you have and what you need.

Innovation 3: Conversational Profile Building — Instead of 20-field forms, AI asks 5-7 natural questions in your language. "Aapki umar kitni hai?" "Aap kya kaam karte hain?" — and extracts all data intelligently.

Innovation 4: AI Scheme Comparison — When multiple schemes match, AI compares benefit amounts, document readiness, and application complexity — then recommends which to apply for first and why, personalized to your situation.

List of features offered by the solution

1. AI Conversational Profile Builder

- AI asks 5-7 simple questions in natural language
- Understands Hindi, English, & Hinglish
- Extracts: age, occupation, state, income, category, family size
- No forms – just chat like talking to a friend

2. Document Intelligence Engine (Innovation 1)

- Upload photo of Aadhaar Card or Ration Card
- AI extracts name, age, gender, address via OCR + LLM
- Auto-fills profile – skips question-answer process
- Aadhaar number auto-masked for privacy (XXXX-XXXX-1234)

3. Smart Scheme Matching (RAG + Rules)

- 3-stage AI pipeline: Rule Filter → RAG Vector Search → LLM Reasoning
- Matches against 30+ verified central government schemes
- Eligibility confidence score for each scheme (e.g., 95% match)
- Covers: Agriculture, Health, Education, Housing, Pension, Women & Child, Employment

4. Document Readiness Scoring (Innovation 2)

- Shows: '3/5 documents ready (60%)'
-  Available documents (uploaded or mentioned)
-  Missing documents with WHERE to get, HOW LONG it takes, and TIPS
- Prioritizes schemes with highest readiness (easiest to apply first)

5. AI Scheme Comparison (Innovation 4)

- Compare 2-3 schemes side by side
- Metrics: benefit amount, document readiness, application complexity, time to benefit
- AI generates personalized recommendation: "Apply for PM-KISAN first because..."

6. Voice Input & Output

- Tap microphone and speak your question
- AI responds in text + audio
- Powered by OpenAI Whisper (STT) + gTTS (TTS)

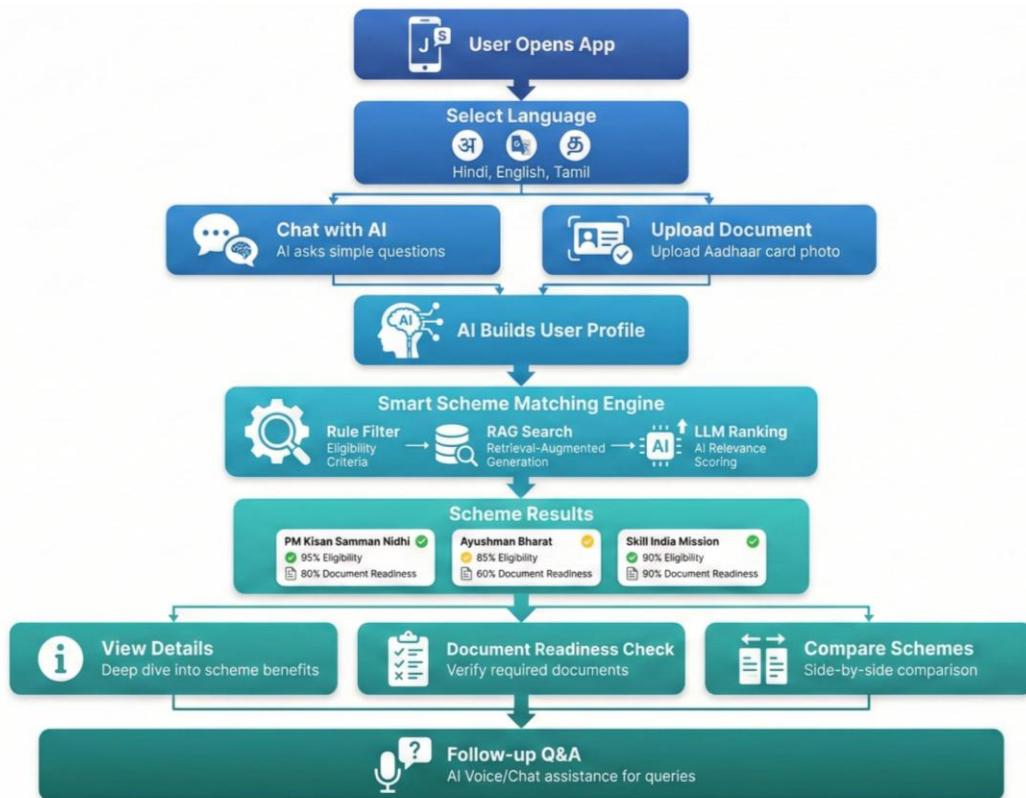
7. Multilingual Support

- Hindi + English (MVP)
- Extensible to Tamil, Telugu, Marathi, Bengali, Kannada, Gujarati, Malayalam, Punjabi, Odia
- Translation via Bhashini API (India's government translation platform)

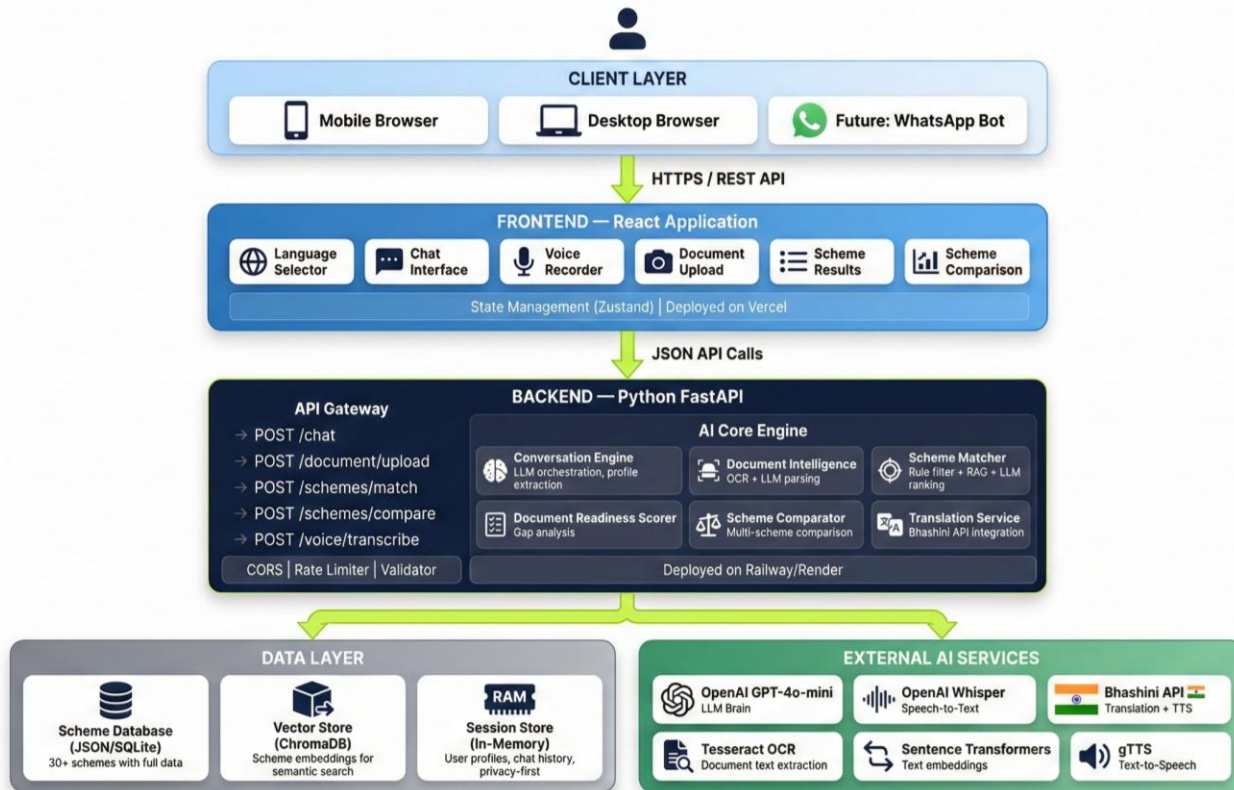
8. Privacy by Design

- Zero persistent data storage – all session-based
- Document images processed in-memory, never saved
- No Aadhaar numbers stored
- No login required

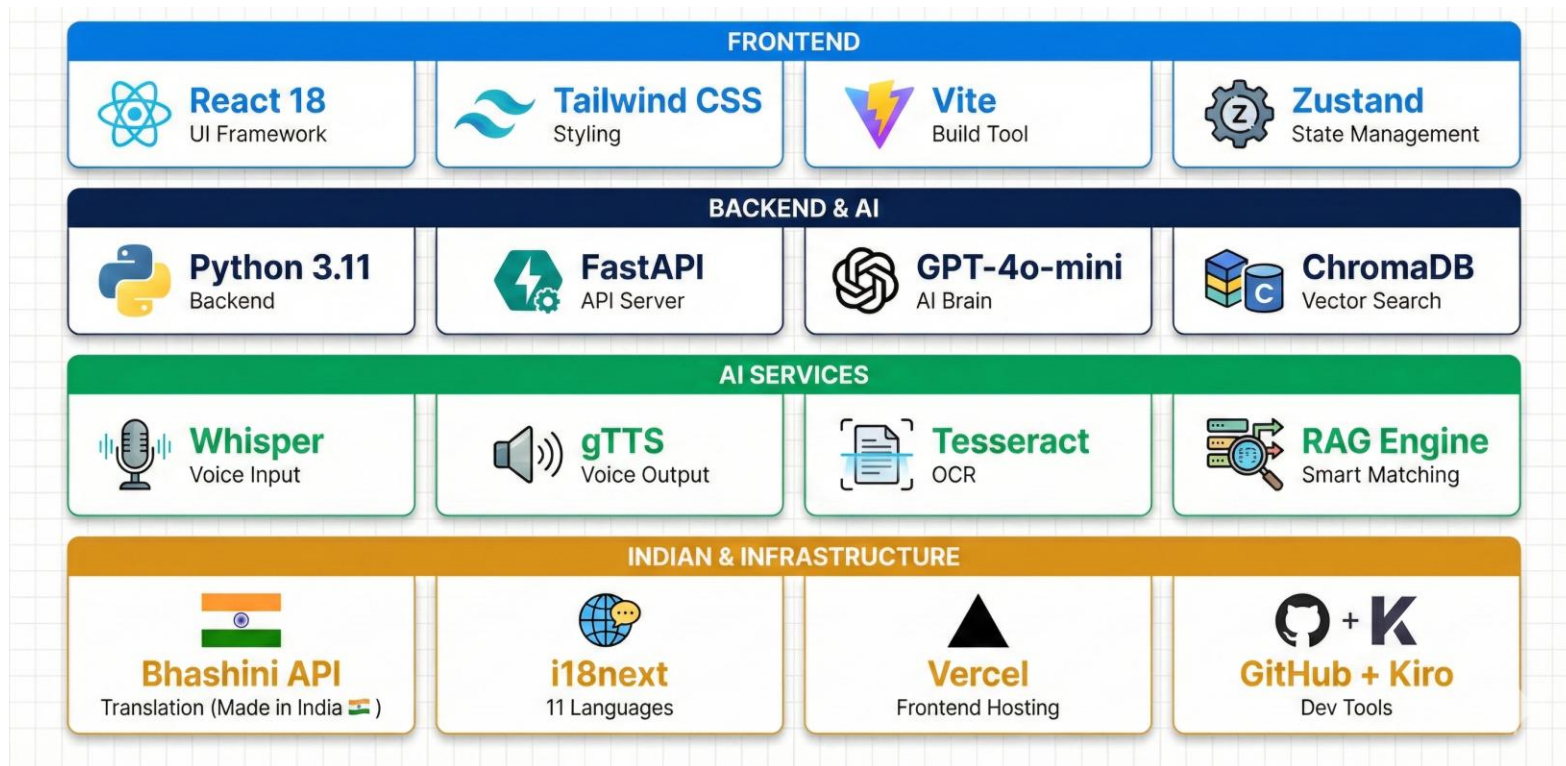
Process flow diagram



System Architecture



Technologies Stack



Estimated implementation cost

Component	Cost	Notes
Vercel (Frontend hosting)	FREE	Community tier
Railway/Render (Backend hosting)	FREE	Free tier (500 hrs/month)
OpenAI GPT-4o-mini	~₹200 for demo	\$0.15/1M input tokens – extremely cheap
OpenAI Whisper	~₹50 for demo	\$0.006/minute
Bhashini API	FREE	Government API, no cost
gTTS	FREE	Open source library
Tesseract OCR	FREE	Open source
ChromaDB	FREE	Open source, embedded
GitHub	FREE	Public repository
TOTAL for Hackathon	~₹250	

Production Scale Cost Estimate:

Scale	Monthly Cost	Per-User Cost
1,000 users/month	₹1,500	₹1.50 per user
10,000 users/month	₹8,000	₹0.80 per user
1,00,000 users/month	₹50,000	₹0.50 per user
10,00,000 users/month	₹3,00,000	₹0.30 per user

ROI Perspective

- Even at ₹3,00,000/month for 10 lakh users, if each user discovers just ONE scheme worth ₹6,000/year, the platform unlocks ₹600 crore in annual welfare benefits – a return of 2,00,000x on technology investment.*
- The cost of NOT building this is ₹5.6 lakh crore in unclaimed welfare every year.*

